

MINAHIL ADIL BUTT

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SISSA/ICTP, Trieste, Italy

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EDUCATION

Masters in High Performance Computing

September 2025- Present

Scuola Internazionale Superiore di Studi Avanzati (SISSA)

International Centre for Theoretical Physics (ICTP)

Supervisors: Prof. Paolo Lazzari, Prof. Riccardo Farneti

Ph.D. in Astrophysics and Cosmology (cum laude)

September 2021- September 2025

Scuola Internazionale Superiore di Studi Avanzati (SISSA)

Supervisors: Prof. Carlo Baccigalupi, Prof. Andrea Lapi, Dr. Sandeep Haridasu

Thesis: "Modified Gravity in Galaxy Clusters: Insights from the Caustic Technique"

Thesis available at: <https://iris.sissa.it/handle/handle/20.500.11767/148210>

Post-graduate diploma programme

September 2020- August 2021

The Abdus Salam International Centre for Theoretical Physics

Supervisor: Prof. Giovanni Villadoro

Thesis: "QCD Axions as a candidate for dark matter"

Bachelors of Science in Physics

August 2016 - July 2020

LUMS University

Supervisor: Prof. Moez Hasan

Thesis: "Hamiltonian Quantization using the ADM formalism"

CGPA : 3.86

The Lyceum School

August 2014 - June 2016

Subjects: Physics, Chemistry, Mathematics

PUBLICATIONS

First Author:

Caustic and hydrostatic mass bias: Implications for modified gravity

Phys. Rev. D 109, 064006

Assessing Modified Gravity via Stacked Galaxy Clusters' Caustics

Preprint: arxiv.org/abs/2504.16685

Other:

Constraining chameleon screening using galaxy cluster dynamics

Phys. Rev. D 108, 044007

Dark Matter in Fractional Gravity III: Dwarf Galaxies Kinematics

Universe 2023, 9(11), 478

Mass Modeling and Kinematics of Galaxy Clusters in Modified Gravity

JCAP 11 (2024) 014

A Relativistic Tensorial Model for Fractional Interaction between Dark Matter and Gravity

Phys. Rev. D 111, 024026

RESEARCH PROJECTS

Polymer Quantum Mechanics

Extensively studied the deviations from usual quantization schemes making use of Spectral Theory of Differential Operators and quantized the Harmonic Oscillator using this scheme.

Project type: Summer Research funded by the **Office of Sponsored Programmes and Research**, under Prof. Moez Hassan, *LUMS*

RELEVANT COURSES

Foundations of HPC
Introduction to Computer Architecture
Advanced Programming (C++)
Parallel Programming
Introduction to GPU Programming
Best Practices for Scientific Computing
Advanced Computer Architecture
Numerical Analysis
HPC Technology
Filesystems and Storage
Advanced GPU Computing
CLOUD for HPC
Introduction to PETSc
Introduction to QC
KS Solvers
Introduction to ML
Introduction to DL
Advanced DL
LLMs and AI Agents
Introduction to RL
Dynamic Workflows

COLLABORATIONS

Member of the Astrophysics and Cosmology group at SISSA (APC group)
Member of the Modified Gravity group at SISSA
Member of the InDark INFN initiative
Enrolled in the Joint ICTP-SISSA PhD Programme
Member of the PAK-ICTP alumni network

PROGRAMMING SKILLS

Languages	MATLAB, Maple, C, C++, Python, Mathematica, Fortran
Softwares	Creo Parametric, LabView, Latex

WORKSHOPS, CONFERENCES AND SEMINARS

Conference- Cosmoverse @ Istanbul	2025
· Contribution: Talk	
Conference- Cosmoverse @ Naples	2025
· Participant	
Conference- IberiCos 2025	2025
· Contribution: Talk	

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| Conference- 59th Rencontres de Moriond
<i>Moriond</i> | 2025 |
| · Contribution: Poster | |
| Conference- 50 years of Horndeski Gravity
<i>Perimeter Institute</i> | 2024 |
| · Contribution: Talk | |
| Conference- CosmoVerse@Krakow
<i>Jagiellonian University</i> | 2024 |
| · Contribution: Poster | |
| Conference- 58th Rencontres de Moriond
<i>Moriond</i> | 2024 |
| · Contribution: Talk | |
| Conference- Cosmology at Miramare 2023
<i>SISSA</i> | 2023 |
| · Contribution: Poster | |
| Workshop on Cosmological tensions and its implications on concordance cosmology
<i>IFPU</i> | 2023 |
| · Participant | |
| School on Future Cosmology
<i>IESC</i> | 2023 |
| · Participant | |
| Workshop on non-perturbative methods in QFT
<i>CERN</i> | 2022 |
| · Attended a two-week workshop bringing together researchers from multiple subfields of theoretical condensed matter and particle physics, focused on the shared goal of solving strongly-coupled systems using non-perturbative methods in QFT. | |
| Seminar on Polymer Quantum Mechanics
<i>LUMS</i> | 2019 |
| · Held a seminar to present my summer research project. | |

TEACHING EXPERIENCE

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| Teaching Assistant for Electricity and Magnetism
<i>LUMS</i> | 2018 |
| · 10hrs/week. Responsible for grading assignments and quizzes and holding office hours. | |
| Teaching Assistant for Modern Physics and Experimental Physics Lab I
<i>LUMS</i> | 2019 |
| · 10hrs/week. Responsible for grading assignments and quizzes and holding office hours. | |
| Teaching Assistant for Quantum Mechanics I
<i>LUMS</i> | 2019 |

- 10hrs/week. Responsible for grading assignments and quizzes and holding office hours.

INSTITUTIONAL RESPONSIBILITIES

Organizer of the Astrophysics Journal Club at SISSA (2022-2023)

Logistics and General Assistance to the PhD initiative- Astrophysics and Cosmology Section (2024-2025)

Library Collaborator at SISSA (2024-2025)

OUTREACH AND VOLUNTEER WORK

Trieste Next

2021

Trieste, Italy

- Volunteered at Trieste Next, a 3-day event aiming to give visibility to research, new technologies, and innovative ideas that contribute to the well-being of communities.

SISSA for Schools

2023-present

SISSA

- Interacted with students from high schools across Italy. Discussed topics in physics and maths. Gave campus tours and shared insights on life in academia.

SISSA Sustainability Club

2024-present

SISSA

- Member of the SISSA Sustainability Club, a student-run group that aims to increase awareness of environmental change and contribute towards mitigating the effects of climate change.

Mashal Academy

2014-2016

The Lyceum School

- Taught math and drama to young students from underprivileged backgrounds.

ACADEMIC ACHIEVEMENTS

Recieved Ph.D. in Astrophysics and Cosmology cum laude

2021-25

SISSA

Placed on the Dean's Honour List

2016-17

LUMS

Placed on the Dean's Honour List

2017-18

LUMS

Placed on the Dean's Honour List

2018-19

LUMS

Placed on the Dean's Honour List

2019-20

LUMS

Won Best Essays Award for research essay on

Music Therapy for Children with Autism

2017

LUMS

Rhodes Scholarship finalist from Pakistan

2019

Graduated with High Distinction
LUMS

2020